

# Introduction to Artificial Intelligence and Machine Learning

TEC-008 · Week 7 of 10 · IT and Technology

Study independently · keep a notebook · save your evidence



## THIS WEEK'S OUTCOME

**Compare risks and benefits of an AI solution.**

## STUDY RHYTHM

**6 hours**

Short sessions work best: Read · Watch · Make · Reflect.

## START HERE · IN PLAIN LANGUAGE

In plain language, this lesson helps you work on compare risks and benefits of an AI solution.. Treat responsible implementation and testing as a skill you can build in small steps: study one idea, compare it with a real Liberian or regional example, and save a short piece of evidence of what you understood.

**01**

**READ**

Find the main idea.

**02**

**WATCH**

Notice a method.

**03**

**MAKE**

Try it locally.

**04**

**REFLECT**

Check what changed.

## Your learning path

### READ FOR THE IDEA

Read for the main idea: Compare risks and benefits of an AI solution.

### WATCH FOR THE METHOD

Watch for one method or example connected to responsible implementation and testing.

### MAKE EVIDENCE

Apply the idea locally and save problem brief and evaluation log.

## ESSENTIAL QUESTION

**How could responsible implementation and testing improve a real situation connected to Introduction to Artificial Intelligence and Machine Learning?**

Write one sentence before you begin. Revisit it after you apply the activity.

## Detailed lesson notes

### CONCEPT NOTE

This original lecture note places “Compare risks and benefits of an AI solution.” inside the wider work of Introduction to Artificial Intelligence and Machine Learning. The week is not only about remembering a definition. It is about understanding the choices behind the work: clear requirements, repeatable workflow, testing, and safe implementation. Begin with a workplace or service process that could be improved with a digital tool, then use the linked academic source to compare your first idea with a more formal explanation. A useful way to learn this topic is to move from a situation to a decision and then to evidence. First describe what is happening and who needs a better result. Next identify the information, people, tools, or constraints that shape the decision. Finally make a small, responsible attempt and explain what you would improve. This sequence keeps self-study practical and prevents a learner from copying a solution without understanding the reason for it.

#### 01 · PURPOSE

Purpose: connect responsible implementation and testing to the outcome “Compare risks and benefits of an AI solution.”.

#### 02 · METHOD

Method: define the user, build a small version, test it with safe sample data, and document the result.

#### 03 · EVIDENCE

Evidence: save a short record that another learner could understand and review.

### INSTRUCTIONS · WORK IN ORDER

1. Read the guide and linked source for the main concept; write down two unfamiliar terms.
2. Describe a local or regional example without including private or identifying information.
3. Complete the weekly exercise using safe sample data, a fictional scenario, or your own non-sensitive work.
4. Review your result against the self-check questions and record one improvement for the next session.

### WEEKLY EXERCISE

Exercise: Introduction to Artificial Intelligence and Machine Learning learners should define the user, build a small version, test it with safe sample data, and document the result. Use the week’s focus as your test case and save problem brief and evaluation log.

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**ACADEMIC READING · SOURCE RECORD**

## Artificial Intelligence

|                  |                                                                       |
|------------------|-----------------------------------------------------------------------|
| Provider         | MIT OpenCourseWare                                                    |
| Kind             | University open course                                                |
| Licence / access | MIT OpenCourseWare terms                                              |
| Official URL     | <a href="https://ocw.mit.edu">Open official source at ocw.mit.edu</a> |

**Use it this week:** Read selectively for this week's responsible implementation and testing work.

**UNIVERSITY OR PROFESSOR VIDEO · SOURCE RECORD**

## Machine Learning Lecture 1

|                  |                                                                                               |
|------------------|-----------------------------------------------------------------------------------------------|
| Provider         | Stanford CS229 · Andrew Ng                                                                    |
| Kind             | University lecture or official educational video                                              |
| Licence / access | Access terms apply; video remains the property of its provider and platform.                  |
| Official URL     | <a href="https://www.youtube.com/watch?v=jNfEjH01S00">Open official source at youtube.com</a> |

**Use it this week:** Watch with the Introduction to Artificial Intelligence and Machine Learning learning outcome in mind; take notes on an idea to test locally.

## Turn study into skill

**PRACTICE LABORATORY**

Complete a guided responsible implementation and testing activity using a Liberian or regional case study.

Keep your work simple, local, and real. A basic note, recording, checklist, screenshot, calculation, or plan is valid practice.

**EVIDENCE CHECKLIST**

- ☐ I named the main idea.
- ☐ I noted one example.
- ☐ I tried one small action.
- ☐ I saved problem brief and evaluation log.

**SELF-CHECK · BEFORE YOU FINISH**

1. What is the main idea in your own words?
2. Which decision or assumption most affects the result?
3. What evidence would show that your approach is useful and responsible?
4. What is one next action I can test or improve?

**REFLECT AND CONNECT**

In 3–5 sentences, explain how this week's concept could improve a real organisation, community service, workplace, or learner project. Identify one assumption, one risk, and one action you would test next.

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# Vocabulary + notebook

Use this final page as a learning checkpoint. Clear terms and brief notes make independent study easier to continue.

## SMALL SELF-STUDY GLOSSARY

**self-directed study (self-study)** means you manage your own learning plan. **learning evidence (evidence)** is the proof you save. A **learning source (source)** is credited to its original provider.

Glossary reference list

**evidence – learning evidence:** A saved note, plan, calculation, recording, checklist, or other item that shows what you can do.

**self-study – self-directed study:** A learner plans, studies, practises, and checks understanding without waiting for a live class.

**source – learning source:** An external reading or video credited to its original provider.

## MY NOTE BANK

The one idea I want to remember:

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The example I will look for this week:

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## FINISH WELL

Choose a time for your next study session. Small regular sessions are more useful than waiting for a perfect long study day.

## SHARE SAFELY

Discuss an idea with a trusted peer, mentor, or colleague. Keep your own notes and do not share sensitive personal or workplace data.

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